Linear Algebra
Assignment #2

1. The solutions must be written in corresponding areas of the assignment.
2. Your solutions must be written legibly and contain logical mathematical reasoning wherever applicable

Degree/Syn: DE-40 EE-A Name: .... Reg. Number: ....

1. Consider the system of equations:

$x + y + z - 2t - s = 6$

$x + 2y - z + 3t = 1$

$2x - y + z - 5t + 2s = 6$

$3x + y - 2z + 2t + 4s = 1$

$x - z + t + 2s = -1$

(a) Write the above system in matrix form:

(b) Write the augmented matrix of the system:

(c) By applying necessary row operations, bring the augmented matrix $A = (a_{ij})$ of the system to the form, where $a_{ij} = 0$ for $i > j$. Note: you must write out every individual step of matrix transformations, indicating the rows being transformed (and how they are being transformed). Proceed as in the examples worked during lectures and exercises.

(d) By looking at your result, explain why the determinant of the system is (obviously) equal to zero, so the system is not a Cramer system.

(e) Write the result of (c) as a new system of equations (equivalent to the original system).

(f) In the last system assume x,y,z for the unknowns and t,s for parameters. Express the variables x,y by t and s (the answer for z below is given to help you correct your possible computational errors).

$z = 3 + 2t + s$

2. Write out the set S of all solutions of the original system
3. Write solution corresponding to $t = 1$ and $s = 0$ and corresponding to $t = 0$ *and* $s = 1$